綴じ代（ホッチキス または 糊付け）

提出期限	年 月 日			受理者
受理日	年 月 日			
評価割合	1	1/2	0	

応用理工物理学実験 レポート

テーマ No.	実験課題
I	Borda's pendulum Experiment Day 1: 1) Measuring the amplitude dependence of period T 2) Measurement of gravitational acceleration Day 2: Computer Simulation: Exact solution, Runge-Kutta method, Numerical calculation of the pendulum equation of motion

実験実施月日	第 1 回	第 2 回
	05 月 12 日	05 月 19 日

N 班	理工学群 応用理工学類 02 年 クラス 202318030 番
	氏名 Andrea Trevanna

共同実験者（必ず記入すること）			
理工学群 応用理工学類 02 年	クラス 202318028 番	氏名	Shaashwath Sankuresh
理工学群 応用理工学類 02 年	クラス 202318030 番	氏名	Nguyen Nguyen Coong
理工学群 応用理工学類 02 年	クラス 202318027 番	氏名	Satish Yash Juriani
理工学群 応用理工学類 年	クラス 番	氏名	
理工学群 応用理工学類 年	クラス 番	氏名	

*このグループにはみんなは総合工学位プログラム（IDE）です

筑波大学 理工学群 応用理工学類

I Borda's Pendulum

Fundamental Labs II Report

Name: Andrea Trevanna Student ID: 202318030

Group: N

1. **Purpose:**

A. **Day 1**

Experiment 1: To gain an understanding of error estimation based on the traditional theme of pendulum vibration: the relations of comparing two types of errors when comparing the experimental and theoretical values, one is based on the measurement and the other is based on theoretical approximation.

Experiment 2: To compare how close we can obtain the value of gravitational acceleration g through experiment by ourselves to the recognised value of gravitational acceleration in Tsukuba which is obtained with the highest possible accuracy by considering a pendulum cycle, maximising its experimental sense.

B. **Day 2**

To solve the problem of pendulum vibration by numerical calculation and estimate the numerical error in the computer experiment. We also compare the obtained result with the experimental results obtained on the first day.
(For the numerical calculation, our team did the Runge-Kutta method because we did only tasks 1, 4, 7, and 8 in our labs.)

1. Purpose

A good experiment should aim to minimize errors (systematic and contingent errors) as much as possible and estimate the error accurately. In addition, it is important to obtain a theoretical formula for the physical model through numerical calculations and compare it with the experimental results.

On the first day, we will provide training on error estimation based on the traditional theme of pendulum vibration. There are two types of errors observed when comparing the experimental and theoretical values, among which one is based on the measurement and the other is based on theoretical approximation. We initially gain an understanding of these relations in Experiment 1. In Experiment 2, the value of gravitational acceleration in Tsukuba is obtained with the highest possible accuracy by considering a pendulum cycle, maximizing its experimental sense. Take the challenge and see how close a value you can obtain with respect to the recognized value of g.

On the second day, we solve the problem of pendulum vibration by numerical calculation and estimate the numerical error in the computer experiment. We also compare the obtained result with the experimental results obtained on the first day.

2. Principle

As shown in Fig. 1-1, we consider a pendulum motion with a sphere of mass m and radius a attached to one end of a thin wire of length l. When the pendulum and hanging string move together, the weight rotates with the vibration of the pendulum. Therefore, the pendulum and entire weight must be treated as a rigid body if the moment of inertia of the weight cannot be ignored.

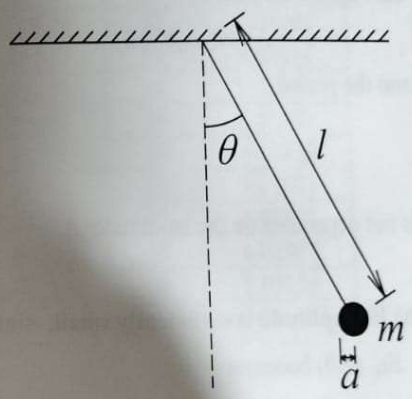


Figure 1-1 Pendulum motion

If the length from the fulcrum of the pendulum (knife edge) to the center of the weight is l, the moment of inertia around the axis passing through the center of the weight is

$$I_0 = \frac{2}{5}ma^2. \tag{2.1}$$

Therefore, the rotational moment around the fulcrum of the pendulum can be given as follows:

$$I = I_0 + ml^2 = ml^2\left\{1 + \frac{2}{5}\left(\frac{a}{l}\right)^2\right\} \tag{2.2}$$

The equation of motion that considers the frictional force $\gamma l\dot{\theta}$ proportional to the velocity is

$$I\ddot{\theta} = -mgl\sin\theta - \gamma l^2\dot{\theta} \tag{2.3}$$

$$\ddot{\theta} = -\frac{\frac{g}{l}\sin\theta + \frac{\gamma}{m}\dot{\theta}}{1+\frac{2}{5}\left(\frac{a}{l}\right)^2} \tag{2.4}$$

Here, if we say,

$$\omega_0 = \sqrt{\frac{g}{l}} \tag{2.5}$$

$$\sigma = \frac{\gamma}{2m} \tag{2.6}$$

this becomes

$$\ddot{\theta} = -\left(\omega_0^2 \sin\theta + 2\sigma\dot{\theta}\right) / \left\{1 + \frac{2}{5}\left(\frac{a}{l}\right)^2\right\} \tag{2.7}$$

In equation (2.7), if we consider that

$$\frac{a}{l} \ll 1 \tag{2.8}$$

the denominator is 1, and Eq. (2.7) becomes Eq. (2.9).

$$\ddot{\theta} + 2\sigma\dot{\theta} + \omega_0^2 \sin\theta = 0 \tag{2.9}$$

As it is difficult to solve this equation, we will try to find approximate solutions for several cases.

(a) If amplitude is sufficiently small, $\sin\theta$ can be approximated to θ and frictional force can be ignored

Equation (2.9) becomes the well-known equation of motion for simple vibration.

$$\ddot{\theta} + \omega_0^2\theta = 0 \tag{2.10}$$

This solution becomes,

$$\theta = A\cos(\omega_0 t + \delta) \tag{2.11}$$

and the period

$$T_0 = \frac{2\pi}{\omega_0} \tag{2.12}$$

is not dependent on the amplitude A.

(b) If amplitude is sufficiently small, $\sin\theta$ can be approximated to θ, but frictional force cannot be ignored

Eq. (2.9) becomes

$$\ddot{\theta} + 2\sigma\dot{\theta} + \omega_0^2\theta = 0$$

This solution, when $\sigma < \omega_0$, with

$$\omega^2 = \omega_0^2 - \sigma^2 \tag{2.13}$$

$$\theta = Ae^{-\sigma t}\cos(\omega t + \delta) \tag{2.14}$$

Such movement is referred to as damped oscillation. The amplitude decreases exponentially, and although period T is constant, it is longer than when there is no friction. That is, (2.15)

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\omega_0^2 - \sigma^2}} \tag{2.16}$$

where you get $T \cong T_0$ is if σ sufficiently small compared with ω_0.

(c) If $\sin\theta$ cannot be approximated to θ, but the frictional force can be ignored

Eq. (2.9) becomes

$$\ddot{\theta} + \omega_0^2 \sin\theta = 0$$

If we apply $\dot{\theta}$ to both sides, and this is integrated using the initial conditions "$\theta = \alpha$ and $\dot{\theta} = 0$ "(α is the maximum swing angle), we get

$$\dot{\theta}^2 = \frac{4g}{l}\left(\sin^2\frac{\alpha}{2} - \sin^2\frac{\theta}{2}\right) \tag{2.18}$$

Here, given

$$k = \sin\frac{\alpha}{2}, \quad \sin\frac{\theta}{2} = k\sin x \tag{2.19}$$

the solution of Eq. (2.17) is

$$\sqrt{\frac{g}{l}}(t - t_0) = \int_0^x \frac{1}{\sqrt{1-k^2\sin^2 x}}dx \tag{2.20}$$

The right-hand side is referred to as a Type 1 elliptic integral and cannot be expressed as an elementary function. Period T is four times the time taken for θ to change 0 to α (x is from 0 to $\pi/2$). Therefore,

$$T = 4\sqrt{\frac{l}{g}}\int_0^{\pi/2} \frac{1}{\sqrt{1-k^2\sin^2 x}}dx \tag{2.21}$$

Table 1-1 shows the relationship between swing angle α and the period ratio T/T_0.

Table 1-1. Relationship between the swing angle α and the period ratio T/T_0

Swing angle (degree)	$\frac{T}{T_0}$	Swing angle (degree)	$\frac{T}{T_0}$	Swing angle (degree)	$\frac{T}{T_0}$	Swing angle (degree)	$\frac{T}{T_0}$
1	1.00002	11	1.00231	21	1.00846	31	1.01861
2	1.00008	12	1.00275	22	1.00929	32	1.01985
3	1.00017	13	1.00323	23	1.01017	33	1.02114
4	1.00031	14	1.00374	24	1.01108	34	1.02246
5	1.00048	15	1.00431	25	1.01203	35	1.02383
6	1.00069	16	1.00491	26	1.01302	36	1.02525
7	1.00093	17	1.00553	27	1.01406	37	1.02671
8	1.00122	18	1.00621	28	1.01513	38	1.02821
9	1.00154	19	1.00692	29	1.01625	39	1.02975
10	1.00191	20	1.00767	30	1.01741	40	1.03134

When k is small, Eq. (2.21) can be expanded as

$$T = 2\pi\sqrt{\frac{l}{g}}\left\{1 + \frac{1}{4}k^2 + \cdots + \prod_{r=0}^{n}\left(\frac{2r-1}{2r+2}k^r\right)^2 + \cdots\right\} \tag{2.22}$$

As k is a function of α, we can see that period T is dependent on the maximum swing angle α.

When limited to $\alpha \to 0$,

$$T \to \frac{2\pi}{\omega_0} = T_0 \tag{2.23}$$

and this corresponds to the case of harmonic oscillation.

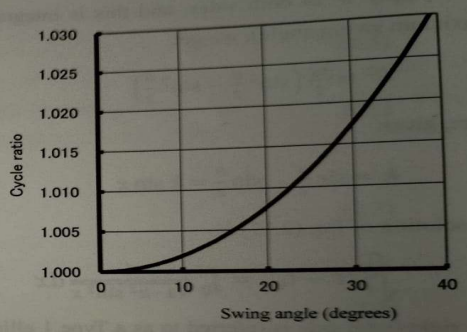


Fig. 1-2. Relationship between swing angle α and the period ratio T/T_0

3. Experiment (Day 1)

3.1 Experiment equipment

Figure 1-3 is a schematic diagram of the experimental equipment. Two types of pendulums are depicted in the figure and both will be used in this experiment.

Parts

- Pendulum weight: Small iron ball with a diameter of approximately 4 cm
- Pendulum suspension wire:
  Approximately 2 m (for micro vibrations)
  Approximately 0.7 m (for large amplitude)

Measurement tools

- Calipers
- Ruler
- Stopwatch

A caliper is used to precisely measure the length of a short object accurately, and the ruler is used to measure a long object. Here, the caliper is used to measure the outer diameter D of the bearing and the radius R of the weight, whereas the ruler is used to measure the length of the suspension wire.

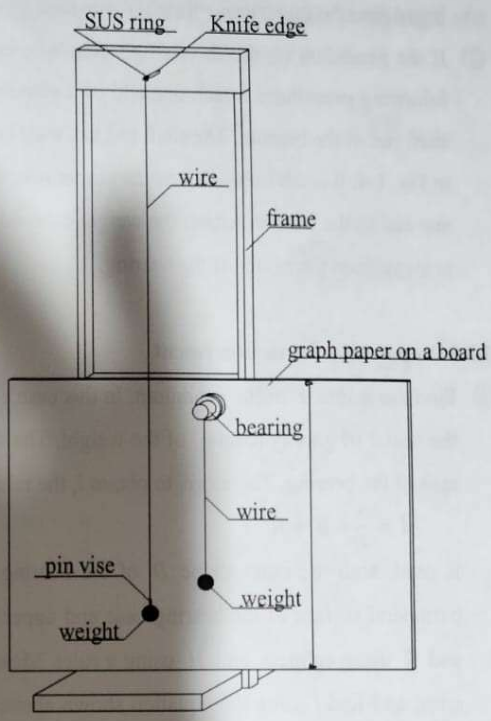


Fig. 1-3 Experimental equipment

3.2 **Experiment 1: Measuring the amplitude dependence of period T**

As described in Section 2(c), when the amplitude of the vibration increases, the vibration deviates from a simple vibration. The purpose of this experiment is to verify this theory empirically. According to the calculated values given in Table 1-1, when the maximum swing angle is 40 degrees, the extent of the relative value deviating from when the cycle has micro vibrations is about 3 %, and so in order to verify the theory, it is necessary to keep the relative error of the measured values in the cycle to approximately 1 % or less.

[Error factors]

When comparing the measured period with values at a small amplitude or approximate calculation value at a small amplitude, the conceivable errors are

(a) Cycle measurement errors ΔT

(b) Length measurement error Δl

(c) By theoretical approximation

(c-1) Error treating weights as a material point, rather than as a rigid body

(c-2) Cycle shifting due to frictional force: If the amplitude is not halved even after 100 vibrations, this can be ignored. (See Appendix 1 of this chapter)

(c-3) Insufficient accuracy of constants: π, g

3.3.1 Preparation for the experiment

① The long suspension wire pendulum will not be used for a while, so where set, move the weight to the back side of the board. Stick the graph paper on to the board (use the graph paper from the previous group where it is still usable).

② If the pendulum on the short suspension wire is not set, set it according to the following procedure. Attach one end of a suspension wire of about 0.7 m to the shaft rod of the bearing. The shaft rod has wide holes and narrow holes, as shown in Fig. 1-4. It is advisable to pass the suspension wire through the shaft rod and tie one end to the washer. Attach the weight especially firmly at a position about 0.6 m away from the center of the bearing.

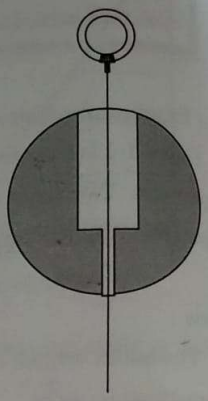


Figure 1-4 Bearing

3.3.2 Preparation for measurement

① Find the length l of the pendulum. In this case, l is the distance from the center of rotation of the bearing to the center of gravity (center) of the weight. The shaft rod rotates around its central axis within the cylindrical case of the bearing. Therefore, to obtain l, the relational expression

$$l = \frac{D}{2} + H + R \tag{3.1}$$

is used, with the outer shape D of the bearing case, length H of the suspension wire between the outer peripheral surface of the bearing case and upper end of the weight, and the weight radius of R. Measure D and R using calipers, and H using a ruler. Measure them 5 times each, find the average value and standard error, and find l using the equation shown above. In order to suppress the relative measurement error of the period to 1 % or less,

$$\frac{\Delta l}{l} = \frac{0.5\Delta D + \Delta H + \Delta R}{l} < 0.01 \tag{3.2}$$

must be satisfied.

② Practice vibrating with α at approximately 40 degrees. Practice so that you can vibrate it 5 times or more without hitting the board. Draw a vertical line on the graph paper at the lowest point and measure the time when the suspension wire crosses it.

③ Estimate the measurement error α. Vibrate for 5 cycles at approximately 40 degrees and measure the time. Repeat the measurement 10 times, recording the results in a notebook, and calculate the mean and standard error. The time measurement error for one cycle is estimated to be about 1/5 of the calculated standard error (when measuring the time for five cycles). If the relative error obtained by dividing this error by the average value is greater than the required accuracy of 1 %, determine the measurement cycle so that the number of cycles to be measured at one time is increased to more than 5, and the relative error of the time measurement of one cycle is smaller than 1 %.

3.3.3 Period Measurement

① Vibrate the pendulum the required number of cycles and measure the time. For the time of one cycle, the measured time may be divided by the number of cycles. Measure α changing the angle to 5 degrees, 10 degrees, 20 degrees, 30 degrees, and 40 degrees respectively, and find the time of one cycle and its standard error. Measurement was performed at 40 degrees when preparing for the measurement, but this measurement should be performed again (under the same measurement conditions as for other angles).

② Draw a graph with α on the horizontal axis and T on the vertical axis. At that time, write the ± standard error as an error bar (See Fig. 1-5). The graph should be attached on the report.

③ Calculate the theoretical value of the period at a small angle and record this at the point of the swing angle of 0 degrees on the graph. In addition, in Tsukuba, the g value is as follows.

$$T_0 = 2\pi\sqrt{\frac{l}{g}} \quad (3.3)$$

$g_{Tsukuba} = 9.7995109\ \mathrm{m/s^2}$

(Latitude 36 degrees 6.0 minutes, longitude 140 degrees 5.5 minutes, height 21.89 m) (3.4)

④ Plot theoretical value of $\alpha = 5, 10, 20, 30, 40$ degree, respectively, by using the T_0 and the table 1-1.

3.3.3 Measurement result

We measured the experimental values through Excel.

As 3.2.2. (1), we measured D/2, H, R, and l five times.

These are the obtained values.

	D/2	H	R	L	
1	23.375	663	19.025	705.4	0.000454
2	23.925	663	19	705.925	-0.00029
3	23.975	663	19.025	706	-0.0004
4	23.125	663	19.025	705.15	0.000808
5	23.175	664	18.95	706.125	-0.00057
avg	23.515	663.2	19.005	705.72	

With the average value of l = 705.72 mm

For the period measurement, we did 4 cycles and change the swing angle from 5 degrees, 10 degrees, 30 degrees, and 40 degrees. We repeated the measurement for each angle 10 times

We then obtained the measurement period values which is the average of the measurements divided by 4 cycles for each respective swing angle. Each measurement method has a relative measurement error of the period of being less than 0.01 (Using the formula given in Equation 3.2)

The measured values is given in the average values recorded in Excel in next page and in Graph 1-1 which is attached to this report. The relationship is period (seconds) with respect to swing angle (in degrees)

The error bars in Graph 1-1 is obtained by having each of the period values (average) ± standard error (std error).

For each swing angle,

Relative error = $\frac{average\ value}{standard\ error}$

	5 degree	10 degree	20 degree	30 degree	40 degree
	6.75	6.85	6.73	6.85	6.91
	6.63	6.73	6.73	6.85	7.17
	6.78	6.65	6.8	6.86	6.86
	6.7	6.69	6.8	6.76	6.9
	6.76	6.76	6.89	6.88	6.88
	6.75	6.75	6.83	6.82	6.83
	6.73	6.63	6.92	6.91	6.97
	6.83	6.73	6.78	6.84	6.93
	6.66	6.87	6.85	6.83	6.95
	6.74	6.79	6.92	6.85	6.93
avg	1.68325	1.68625	1.70625	1.71125	1.73325
std error	0.004565	0.00616	0.005543	0.003101	0.00737
relative error	0.002712	0.003653	0.003249	0.001812	0.004252
std(-)	1.678685	1.68009	1.700707	1.708149	1.72588
std(+)	1.687815	1.69241	1.711793	1.714351	1.74062

For the theoretical value of the period at a small angle:

From the equation,

$T_0 = 2\pi\sqrt{\frac{l}{g}}$ =1.68614097 at swing angle 0 degrees

With g as $g_{Tsukuba}$ = 9.7995109 m/ s^2

Using T_0 and Table 1-1, we then obtained the theoretical values of swing angle α = 5, 10, 20, 30, and 40 degrees

T at α = 5 degrees = $T_0 \times 1.00048$ T at α = 10 degrees = $T_0 \times 1.000191$

T at α = 20 degrees = $T_0 \times 1.00048$ T at α = 30 degrees = $T_0 \times 1.001741$

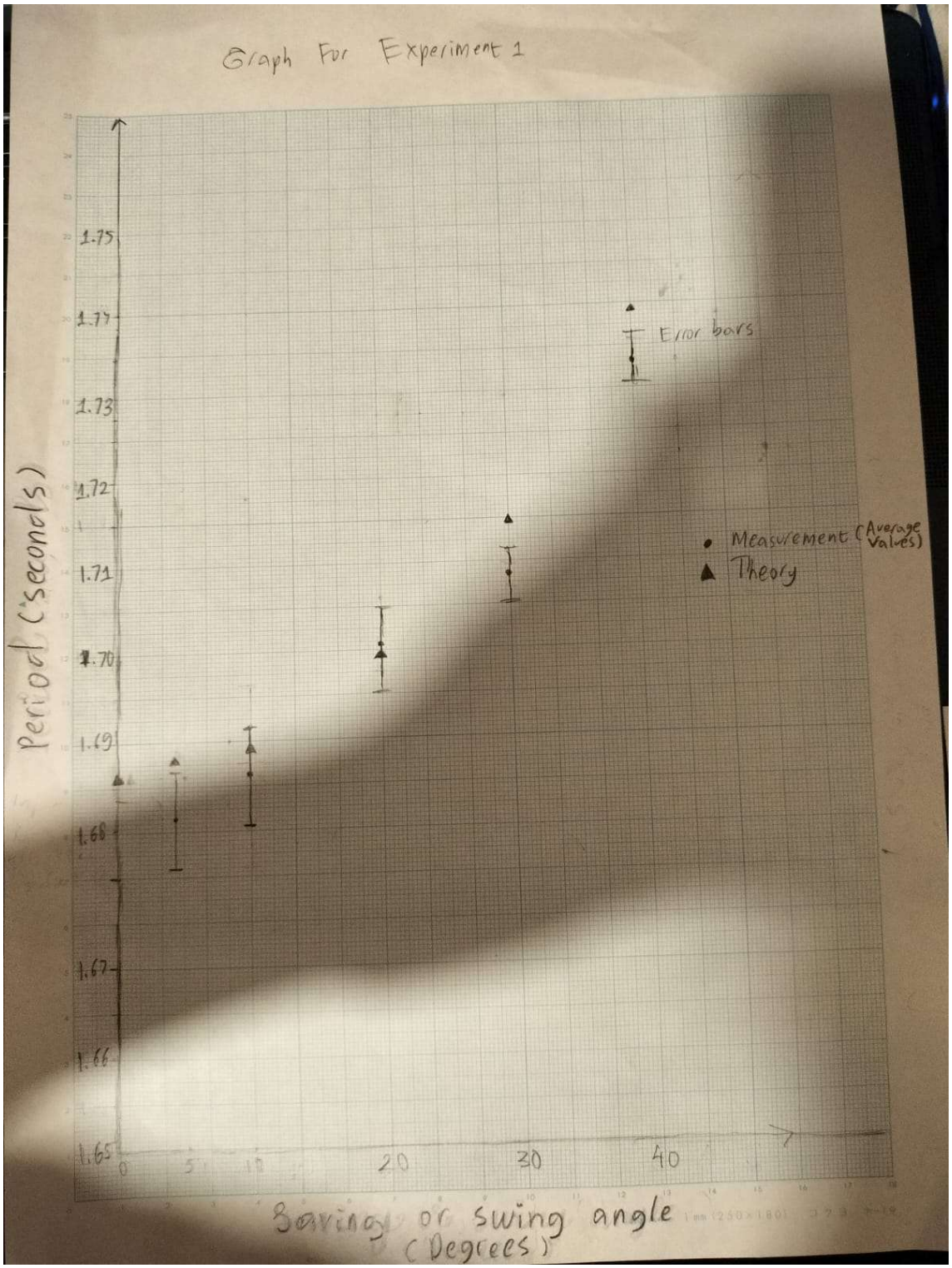
T at α = 40 degrees = $T_0 \times 1.03134$

From table 1.1						
	Degree	5	10	20	30	40
Theoritical	Time	1.686950315	1.689361497	1.699074	1.715497	1.738985

The theoretical period values are also recorded in Graph Experiment 1 with the triangle symbol

The graph for Experiment 1 is next page

Graph Experiment 1: Expected Graph (Period Swing Angle Dependence)



Swing angle (degrees) = α

3.3.4 Discussion (Report assignments)

1) From the above graph, it can be seen that in general, as the initial swing angle increases, the period, or the time taken to complete one oscillation also increases. Thus, it can be said that the period T is a monotone increasing function of α.

2) T increases with swing angle α due to:

a) Restoring Force Weakens Relative to Displacement:

- As θ increases, sinθ grows more slowly than θ itself.
- This makes the **restoring torque less effective** compared to what it would be under the small-angle assumption.

b) Longer Path + Slower Average Speed:

- The pendulum swings along a **longer arc** for larger α\alphaα.
- But because the restoring force isn't as strong, it **accelerates less**, so the pendulum takes **more time** to complete a cycle.

As for if the restoring force is proportional to the displacement as in Eq(2.10) in Section 2(a), T is a constant regardless of α. Considering the difference from Eq(2.17) in section 2(c):

The reason the period is an increasing function of α is due to the described nature of the motion of a pendulum in equations 2.10 and 2.17. The reason the actual results align with equation 2.17 and not 2.10 is because equation 2.10 assumes an angle $\alpha \approx 0$, which is not true in the case of this experiment for angles greater than 5°. therefore, the period seems to have an increasing trend for these angles as modelled by equation 2.21.

3)

Random Error Characteristics:

Random errors arise from unpredictable fluctuations in measurement, such as inconsistent reaction times when starting/stopping a stopwatch. These errors vary in direction and magnitude across trials. In the pendulum experiment, slight delays or premature stops by the experimenter would cause the measured period to sometimes be longer or shorter than the true value.

Impact on Results:

- Random errors manifest as variability in repeated measurements of the same period. For example, timing 10 oscillations once would reduce the per-period error (e.g., ±0.01 s)4, but repeated manual timing would show scatter due to reaction-time inconsistencies7.
- Averaging multiple measurements can mitigate random errors, as positive and negative deviations cancel out

Contrast with Systematic Error:

- Systematic errors would occur if the stopwatch itself was miscalibrated (e.g., consistently running slow) or if the experimenter **always** delayed stopping the timer by a fixed amoun

However, the question specifies "stopping error" tied to reaction time, which is inherently random. Manual stopwatch use would introduce random variability. In the case of our experiment, we use combination of phone timer and manual stopwatch and often start late , so it is random error.

4) The measured values agree theortically within margin of error for swing degrees $\alpha = 10{,}20$ degrees. However, for other angles, the theoretical values T= $T_0 \times$ $some\ constant\ in\ Table$ 1.1 do not match.

This could be because of frictional force acting on the weight at smaller angles due to a smaller momentum. The frictional force would inhibit motion causing damped oscillation requiring a longer time to complete an oscillation, hence having a larger period than the theoretical value.

5)

In order for it to be permissible to treat the weight as a point mass, either the diameter of it would have to be extremely small or the swing angle must be extremely small. As illustrated in the equation below,

$$\frac{2}{5}\left(\frac{\alpha}{l}\right)^2 < 0.001$$

Using l= 705.72,

$$\alpha < 35.286$$

Which is well inside our experimental values for swing degrees

3.4 Experiment 2: Measurement of gravitational acceleration

To what level of precision can gravitational acceleration be calculated? Mobilize the tests, experimental sense, and observations learned in Experiment 1 and use your own ingenuity to find the most accurate value possible. Here, the amplitude dependence of the period is reduced by decreasing the swing angle and increasing l. In such a case, the error between the relative error of the length measured value and the theoretical formula error when treating the weight as a material point can be reduced. As a result, the theory that should be compared with the experiment is

$$T_0 = 2\pi\sqrt{\frac{l}{g}} \tag{3.7}$$

In this experiment, using Eq. (3.7), we can obtain the value of g from the measured values of l and T_0. At this time, estimate the error associated with the g value. In order to reduce the error of T_0, it is necessary to measure a large number of cycles each time.

3.4.1 Experiment preparation

① Replace the weight that was moved to the back of the board.

② If the pendulum of the long suspension wire is not set, set it according to the following procedure. Tie one end of approximately 2 m of piano wire through the hole of the stainless ring and hang it on the knife edge. Secure the weight using a pin vice approximately 1.8 m from the knife edge.

3.4.2 Measurement preparation

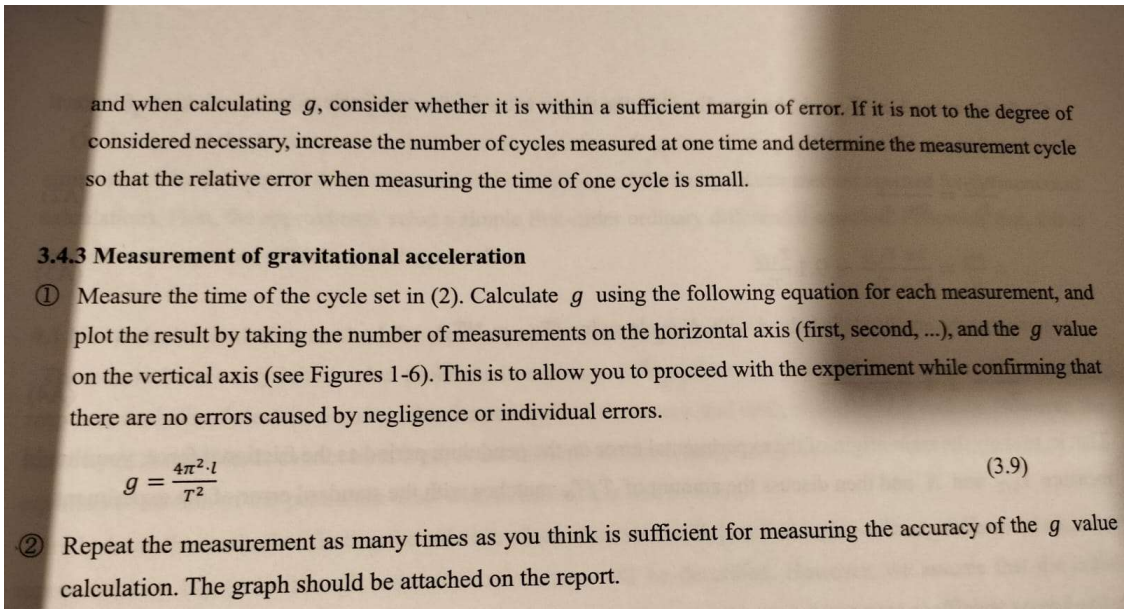
① Measure the diameter of the weight. Measure this for 5 times with a caliper and calculate the radius R based on the average value.

② Measure the length of the pendulum. Using a long measuring instrument installed in the laboratory (a ruler is attached to the bottom of the knife edge), measure the distance l' from the knife edge to the bottom of the weight. l can be obtained from the relational expression,

$$l = l' - R \tag{3.8}$$

Repeat measurement of l' for 5 times, to find l for 5 times, and calculate the error. Even if the values measured 5 times are all the same value, the error is set to 1/2 of the lowest digit of the measured values, due to the measurement accuracy of the equipment used.

③ Set the swing width of the weight to approximately 6 cm (on one side of the vertical line) (approximately twice at swing angle α), and measure the time for 10 cycles using a stopwatch. Repeat this process three times to get a sense of the T value (per cycle) and the measurement error. This also provides practice at swinging the pendulum in a vertical plane and of accurately measuring cycles. Calculate the relative error of T per cycle,



and when calculating g, consider whether it is within a sufficient margin of error. If it is not to the degree of considered necessary, increase the number of cycles measured at one time and determine the measurement cycle so that the relative error when measuring the time of one cycle is small.

3.4.3 Measurement of gravitational acceleration

① Measure the time of the cycle set in (2). Calculate g using the following equation for each measurement, and plot the result by taking the number of measurements on the horizontal axis (first, second, ...), and the g value on the vertical axis (see Figures 1-6). This is to allow you to proceed with the experiment while confirming that there are no errors caused by negligence or individual errors.

$$g = \frac{4\pi^2 \cdot l}{T^2} \qquad (3.9)$$

② Repeat the measurement as many times as you think is sufficient for measuring the accuracy of the g value calculation. The graph should be attached on the report.

We decided to measure the gravitational acceleration 12 times to increase accuracy.

3.4.3 Measurement results

Doing 3.4.2 (1) and (2), we obtained these values for the diameter of the wight R, distance l', and l for 5 times. The average values of R, l', and l are then obtained. (avg=average)

	R	l'	l
	19.625	1755	1735.375
	19.55	1754	1734.45
	19.625	1756	1736.375
	19.65	1755.5	1735.85
	19.65	1756	1736.35
avg	19.62	1755.3	1735.68

After doing 3.4.2 (3) and 3.4.3 (3) to obtain T and the measurement/experimental value of g 12 times, using equation (3.9) $g = \frac{4\pi^2 \cdot l}{T^2}$,

We then obtain the measurement values of g. Each measurement value is recorded in Graph Experiment 2

	T		g	stdError(-)	stdError(+)
1	26.55		9.720762782	9.682569	9.758957
2	26.44		9.80181484	9.763621	9.840009
3	26.7		9.611847531	9.573653	9.650042
4	26.56		9.713444308	9.67525	9.751639
5	26.71		9.604651689	9.566457	9.642846
6	26.76		9.568793397	9.530599	9.606988
7	26.27		9.929085497	9.890891	9.96728
8	26.41		9.824095905	9.785902	9.86229
9	26.36		9.861400203	9.823206	9.899594
10	26.38		9.846453028	9.808259	9.884647
11	26.3		9.90644651	9.868252	9.944641
12	26.46		9.787002882	9.748809	9.825197
avg	2.649167		9.764649881	9.726456	9.802844
std error	0.004745		0.038194274		
relativeErr	0.001791		0.003911484		

The average value of g = 9.764649881 m/s^2 .

The errors are:

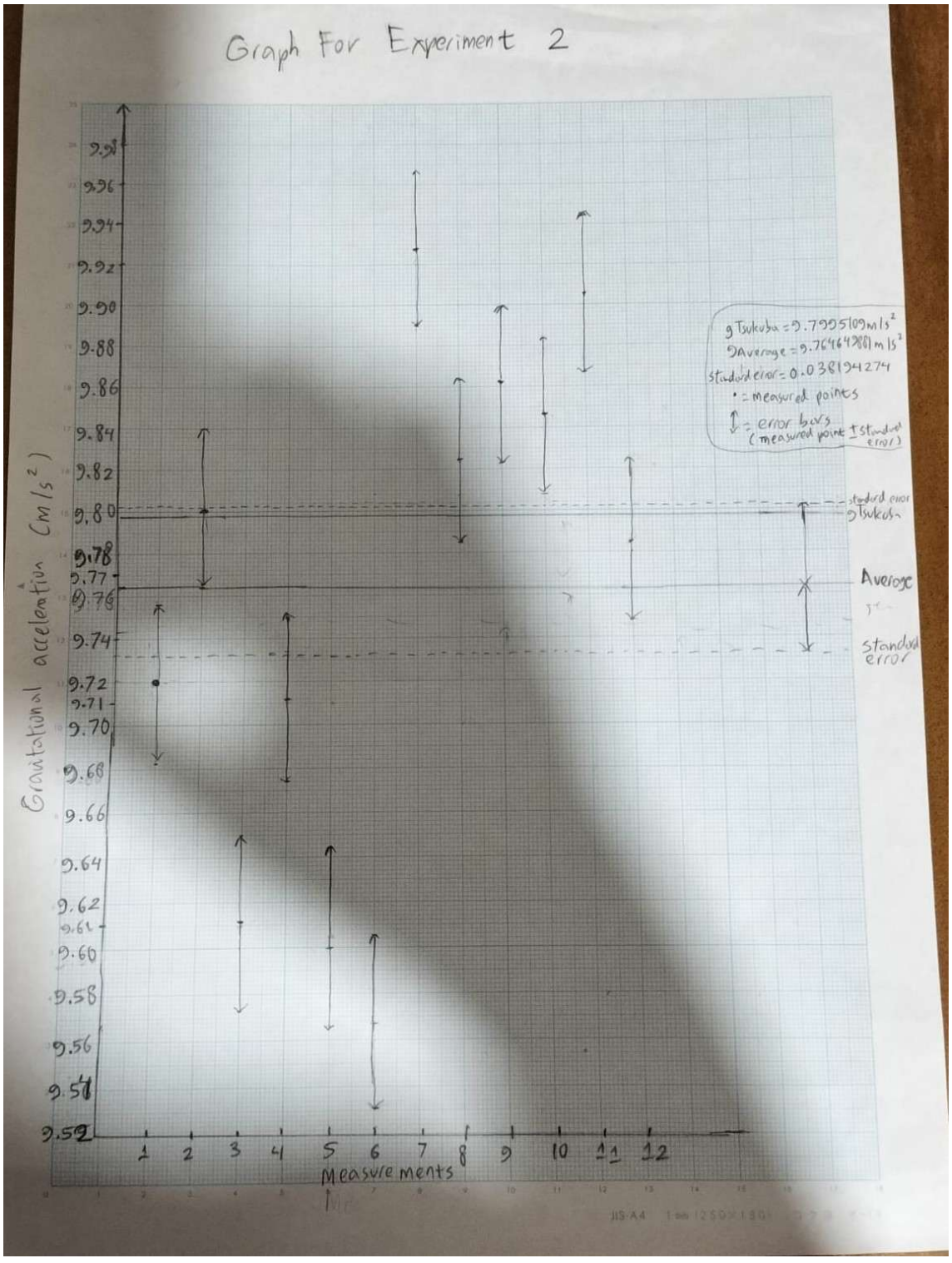
Standard error (for g)= 0.038194274

(Standard error bars in Graph Experiment 2 are at g average $\pm$ standard error)

Relative error = $\frac{average\ value\ of\ g}{standard\ error\ for\ g}$ = 0.003911484

Graph of Experiment 2 is next page

Graph for Experiment 2: Measured value of Gravitational Acceleration



3.4.4 Discussions (report Assigments)

1) The average value of g = 9.764649881 m/s^2

Error of g is:

Standard error (for g)= 0.038194274

Relative error = $\frac{average\ value\ of\ g}{standard\ error\ for\ g}$ = 0.003911484

(Standard error bars in Graph Experiment 2 are at g average $\pm$ standard error)

So the standard error bars are at g = 9.726456 and 9.802844 m/s^2

2) With the value of gTsukuba given in section 3.3.3 as

$g_{Tsukuba} = 9.7995109\ m/s^2$,

The difference between the average of g and gTsukuba = 9.7995109-9.764649881 = 0.034861

The difference between the values is quite small, although our average value for g is lower than gTsukuba. The origin of the difference could be due to human errors in experimentation, such as imperfect stopping of the stopwatch (which was tried to be eliminated by repeating the experiment for 12 trials). The frictional force acting on the pendulum could affect its period, causing a damped oscillation. The cause of this force could be slight wind movement, or even lateral movement of the pendulum (front and back rather than perfectly side-to-side). Another reason could be due to slight fluctuations in the g value depending on the rotation of the earth and altitude.

4. Day 2 Computer Simulation

On the second day, a numerical solution of the equation of motion of the pendulum is obtained by computer simulation. The Euler method, the improved Euler method, and the Runge-Kutta method are used for the numerical calculations. First, the approximate value a simple first-order ordinary differential equation. Following that, this is applied to the equation of motion of the pendulum.

4.1 Numerical calculation method for ordinary differential equations

There have been many initial value problems in ordinary differential equations from long ago, such as transient response analysis of electric circuits, vibration analysis of structures, and orbit calculation of rockets. However, the majority of the problems cannot be solved, or are difficult to solve, analytically, so must be solved numerically. The equation of motion of the pendulum dealt with in this task is one of these.

Firstly, here, the method of finding the solution of the constant coefficient first-order ordinary differential equation represented by Eq. (4.1) through numerical calculation will be described. However, we assume that the initial value$x(t_0) = x_0$ is applied. The pendulum equation used in this experiment is a constant coefficient second-order ordinary differential equation, but, as will be described later, the second-order ordinary differential equation is reduced to a two-dimensional first-order simultaneous ordinary differential equation by replacing variables. It can be treated in the same way as for the first dimension.

$$\dot{x}(t) = f(t, x(t)) \tag{4.1}$$

There are several known methods for solving ordinary differential equations through numerical calculation, but the method used in this experiment is adding time axis from an initial point.$(t_0, x(t_0))$ and finding the value of $x(t)$, in order. That is to say, $(t_{i+1}, x(t_{i+1}))$ is obtained from the points $(t_i, x(t_i))$ $(i = 0,1,2, \dots)$ required at each stage From $(t_{i+1}, x(t_{i+1}))$ as follows.

$$t_{i+1} = t_i + h$$
$$x(t_{i+1}) = x(t_i) + \Delta x(t_i) \tag{4.2}$$

Below, we explain the method of calculating $\Delta x(t_i)$.

4.4 Runge-Kutta method

The Runge-Kutta method is an approximation method similar to the Euler method and the improved Euler method, which uses weighted averages of various calculations to obtain even higher accuracy $\Delta x(t_i)$ approximation. The r- stage Runge-Kutta method is written as follows.

$$\Delta x(t_i) = x(t_i) + a_1 k_1 + a_2 k_2 + \cdots + a_r k_r \tag{4.8}$$

Here, $k_j\ (j = 1,2,\cdots,r)$ consists of various calculated values of $\Delta x(t_i)$, and a_j is a constant taken in a way that Eq. (4.8) matches Eq. (4.5) as much as possible. For example, in case $r = 2$,

$$k_1 = hf(t_i, x(t_i))$$
$$k_2 = hf(t_i + h, x(t_i) + k_1)$$

if $a_1 = a_2 = 1/2$, this becomes the improved Euler method.

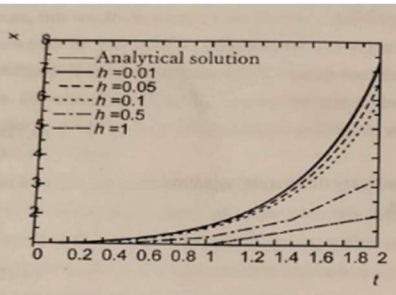


Figure 2-3 Euler method

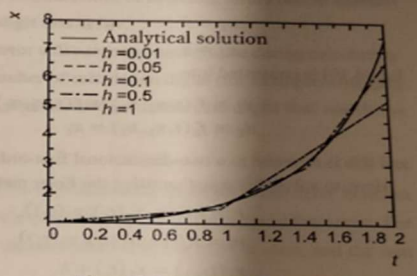


Figure 2-4 Improved Euler method

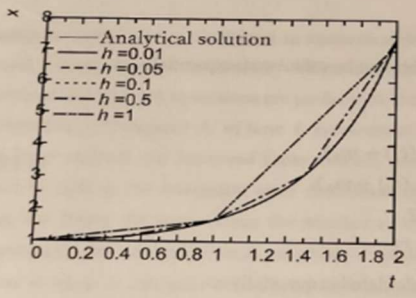


Figure 2-5 Runge-Kutta method

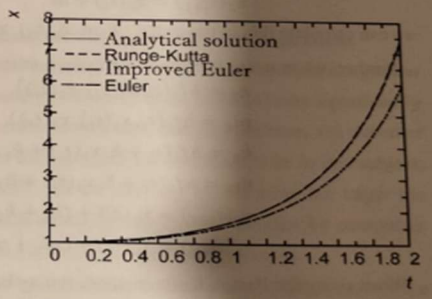


Figure 2-6 Comparison of each method when h = 0.1

Therefore, the improved Euler method can be considered as a two-stage Runge-Kutta method. The method used in this experiment is a classical four-stage method, as shown by the following formula (4.9). This method is widely as it is both easy to calculate and has good accuracy. In general, the 4-stage Runge-Kutta method$O(h^4)$ is known to produce accuracy up to $O(h^4)$.

$$x(t_{i+1}) = x(t_i) + (k_1 + 2k_2 + 2k_3 + k_4)/6$$
$$k_1 = hf(t_i, x(t_i)) \tag{4.9}$$
$$k_2 = hf(t_i + h/2\ , x(t_i) + k_1/2)$$

$$k_3 = hf(t_i + h/2\,, x(t_i) + k_2/2)$$
$$k_4 = hf(t_i + h\,, x(t_i) + k_3)$$

4.5 Constant coefficient second order ordinary differential equations

The discussion so far has been in relation to constant coefficient first order ordinary differential equations. In order to solve the constant coefficient second-order ordinary differential equations such as those handled in this experiment using numerical calculation, the second-order ordinary differential equations need to be transformed into two-dimensional first-order simultaneous ordinary differential equations. Let us say the differential equation to discover is

$$\ddot{x} = f(t, \dot{x}(t), x(t)) \tag{4.10}$$

Here, if we substitute

$$x_1 = \dot{x}$$
$$x_2 = x$$

Eq. (4.10) is converted into

$$\dot{x_1} \equiv f_1(t, x_1, x_2) = f(t, x_1, x_2)$$
$$\dot{x_2} \equiv f_2(t, x_1, x_2) = x_1 \tag{4.11}$$

and this is returned to a two-dimensional first-order ordinary differential equation.

Here, to solve this equation using the Euler method,

$$k = hf_1(t_i, x_1(t_i), x_2(t_i)),$$
$$m = hf_2(t_i, x_1(t_i), x_2(t_i)),$$
$$x_1(t_{i+1}) = x_1(t_i) + k,$$
$$x_2(t_{i+1}) = x_2(t_i) + m \tag{4.12}$$

we can calculate this in the order…$k, m, x_1(t_2), x_2(t_2)$.

When using the Runge-Kutta method, it may be calculated sequentially as

$$k_1 = hf_1(t_i, x_1(t_i), x_2(t_i)),$$
$$m_1 = hf_2(t_i, x_1(t_i), x_2(t_i)),$$
$$k_2 = hf_1(t_i + h/2, x_1(t_i) + k_1/2, x_2(t_i) + m_1/2),$$
$$m_2 = hf_2(t_i + h/2, x_1(t_i) + k_1/2, x_2(t_i) + m_1/2),$$
$$k_3 = hf_1(t_i + h/2, x_1(t_i) + k_2/2, x_2(t_i) + m_2/2),$$
$$m_3 = hf_2(t_i + h/2, x_1(t_i) + k_2/2, x_2(t_i) + m_2/2),$$
$$k_4 = hf_1(t_i + h, x_1(t_i) + k_3, x_2(t_i) + m_3),$$
$$m_4 = hf_2(t_i + h, x_1(t_i) + k_3, x_2(t_i) + m_3),$$
$$x_1(t_{i+1}) = x_1(t_i) + (k_1 + 2k_2 + 2k_3 + k_4)/6, \tag{4.14}$$

$$x_2(t_{i+1}) = x_2(t_i) + (m_1 + 2m_2 + 2m_3 + m_4)/6$$

4.6 Numerical calculation error

The errors in numerical calculations are as follows.

(a) Error due to the floating-point system of the computer

These are computer-specific errors, and consist of rounding errors, information loss errors, and digit loss errors. **Rounding errors** occur when the finite number of digits that can be handled by the computer are exceeded, and due to the lower digits being truncated, an error occurs with the true value. Rounding errors are frequently a problem when calculating a large number of additions and subtractions, and the result of accumulating the respective rounding errors, this results in a huge error. Errors caused by the accumulation of rounding errors due to addition and subtraction may be referred to as **information loss error** or **digit loss error**.

It is often difficult to theoretically predict how much rounding error will occur and accumulate during calculation. In such a case, for example, try making various changes to the number of calculation digits (for example, changing to single precision, double precision, quadruple precision, etc.), and checking whether the digits that match the obtained results are correct.

(b) Error inherent in the calculation algorithm

In addition to the error in (a), there may also be errors specific to the calculation algorithm. The error between the numerical value theoretically obtained by the algorithm and the true value is called the **truncation error**. For example, in the Euler method, the approximate expression up to the second term of Eq. (4.5) is used, and the sum of the third and subsequent terms is the truncation error.

4.7 Determining the error and increment h in numerical solutions for ordinary differential equations

The numerical solution method for ordinary differential equations handled in this task is a numerical integration method, and since many addition operations are performed, the rounding error described in 4.6 (a) may significantly accumulate. Therefore, if increment h of time t is too small, rounding errors can become a problem. On the other hand, with the Euler method, the improved Euler method, and the Runge-Kutta method dealt with in this subject, attention should be paid to the truncation error described in 4.6(b). On the contrary in this case, the larger the increment h is, the larger the error. Since the number of calculation digits that can be handled by computers recently has increased, latter truncation error may have more impact than the former rounding error.

The increment h must be carefully determined for each algorithm, based on the calculation algorithm, and the required level of calculation precision. There are many textbooks and papers that theoretically discuss this method, but unless the essence of the function of f is known, it is often not clear what to do in such a case.

In practical terms, it is good to repeatedly solve the same problem by starting from a large h and gradually decreasing h (for example, halve it, then make it 1/10, and then 1, 0.5, 0.2, 0.1, 0.05, 0.02, 0.01, ... in a ratio of 5: 2: 1) , For example, as the calculation accuracy of the Euler method is $O(h)$, as described in Section 4.2., if you halve h, the error will also halve In other words, the accuracy of $(O(h^4)$, which is "the difference between the solutions to $h = 0.2$ and $h = 0.1$" is approximately is about 1/16 with the Runge-Kutta method). In such a case, by comparing the solution for a different h, the error compared to the true value can be estimated to some extent. Let us attempt to formularize this.

If the truncation error of a numerical solution calculated with increment h is $E(h)$, we can consider a case where

$$E(h) = ah^p \tag{4.15}$$

is formed. Here, a is the factor proportionality, and p is the convergence order of the converged algorithm, which is 1 with the Euler method, 2 with the improved Euler method, and 4 with the Runge-Kutta method.

With increment h_1, h_2, the solution is $x(h_1), x(h_2)$ and a true value of $\bar{x}$,

$$x(h_1) = ah_1^p + \bar{x}$$
$$x(h_2) = ah_2^p + \bar{x}$$

If we use these differences to obtain a,

$$a = \frac{x(h_1)-x(h_2)}{h_1^p - h_2^p}$$

Therefore, the error we want to obtain is

$$E(h_1) = (x(h_1) - x(h_2))\frac{h_1^p}{h_1^p - h_2^p} \tag{4.16}$$

Since all right sides of this equation are all quantities obtained from the calculation results, the error for a particular increment can be estimated using the results from a different increment.

Principle and Preparation

The pendulum equation used in this experiment is a constant-coefficient scond-order ordinary differential equa tion but can be reduced to a two-dimensional first-order simultaneous ordinary differential equation by replacing variables. Hence it can be treated in the same way as first dimension as shown in the equation below,

$\ddot{x}$ (t) = f(t,x(t))

This can be done by the Range-Kutta Method

For preparation as an exercise, calculate the approximate value of the solution of the first-order ordinary differential equation in Eq. (4.14) The exact solution of this ordinary differential equation is x(t) = exp $\frac{t^2}{2}$ is

$$\dot{x}(t) = t \cdot x(t) \equiv f(x,t) \qquad (4.17)$$

$$x(0) = 1 \text{ (initial value)}$$

For numerical calculation programming and outputting into a csv file, we use Spyder using Python language and the numpy module

[Task 1] Exact solution

(1-1)

After running code 4-1 into the csv file 'practice_true.csv', we got the following values:

time	Exact solutior
0	1
0.1	1.005013
0.2	1.020201
0.3	1.046028
0.4	1.083287
0.5	1.133148
0.6	1.197217
0.7	1.277621
0.8	1.377128
0.9	1.499303
1	1.648721
1.1	1.831252
1.2	2.054433
1.3	2.327978
1.4	2.664456
1.5	3.080217
1.6	3.59664
1.7	4.241852
1.8	5.05309
1.9	6.079971
2	7.389056

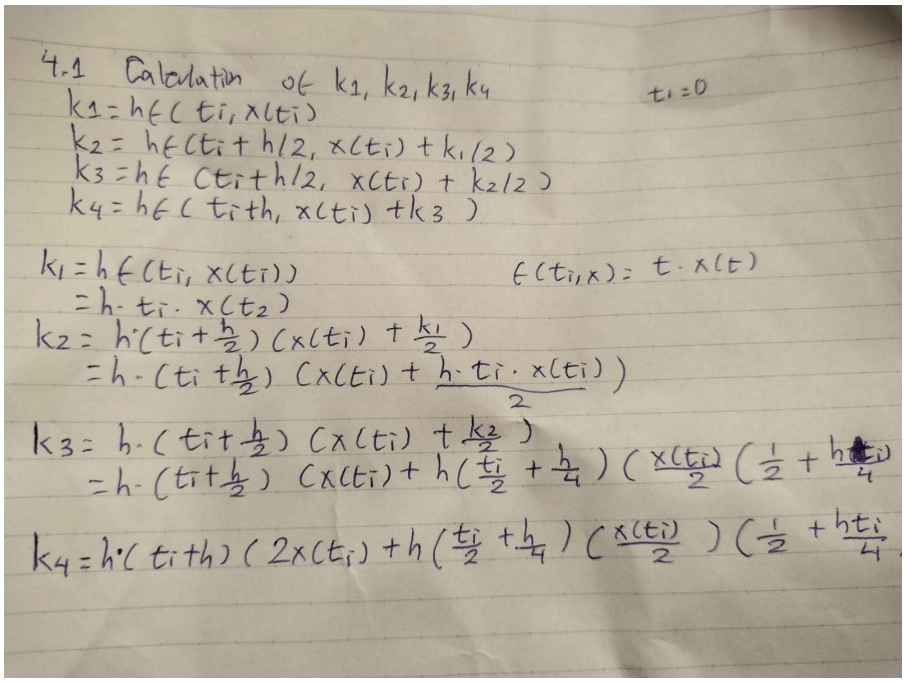
with time in seconds and in time increments of h=0.1 from 0 to 2 s

Task 4

(4-1) Solving the numerical solution of Eq. (4.17) using the Runge-Kutta method:

Substituting Eq. 4-17 into Eq.4-9 and substituting k1, k2, k3, k4 into x(t_i), t_i , h,

We obtain the following values



(4-2) For the Runge-Kutta calculation, the code is quite similar to Code 4-2

```
import numpy as np

#End time [s]
t_end = 2
#Time increment[s]
h = 0.1
#points
N = int(t_end / h + 1)
#Time
t = np.linspace(0, t_end, N)
#x
x = np.linspace(0, t_end, N)

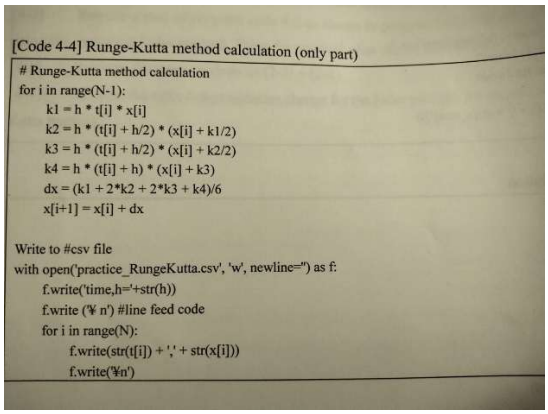
#Initial condition
x[0] = 1
```

Except for the changed part in the bottom:

Changed from the original code into Code 4-4 (only par)

```
#Euler method calculation
for i in range(N-1):
    k = h * t[i] * x[i]
    x[i+1] = x[i] + k

Write to #csv file
with open('practice_Euler1.csv', 'w', newline='') as f:
    f.write('time,h='+str(h))
    f.write ('¥ n') #line feed code
    for i in range(N):
        f.write(str(t[i]) + ',' + str(x[i]))
        f.write('¥n')
```



[Code 4-4] Runge-Kutta method calculation (only part)

```
# Runge-Kutta method calculation
for i in range(N-1):
    k1 = h * t[i] * x[i]
    k2 = h * (t[i] + h/2) * (x[i] + k1/2)
    k3 = h * (t[i] + h/2) * (x[i] + k2/2)
    k4 = h * (t[i] + h) * (x[i] + k3)
    dx = (k1 + 2*k2 + 2*k3 + k4)/6
    x[i+1] = x[i] + dx

Write to #csv file
with open('practice_RungeKutta.csv', 'w', newline='') as f:
    f.write('time,h='+str(h))
    f.write ('¥ n') #line feed code
    for i in range(N):
        f.write(str(t[i]) + ',' + str(x[i]))
        f.write('¥n')
```

When the code was run, we obtained these value for h=0.1and h=0.5 (in summary, here are some of the values)

time	h0.1
0	1
0.5	1.117479
1	1.550367
1.5	2.680192
2	5.79367

time	h0.5
0	1
0.5	1.149434
1	1.699568
1.5	3.290587
2	8.471662

(4-3) The error rounded to 4 digits after decimal place for h=0.1 is below, I used the formula Error= Exact solution – runge kutta values

time	h0.1	Exact solution	Error	Error4digits
0	1	1	0	0
0.1	1.005101	1.005013	−8.83583E−05	−0.0001
0.2	1.018885	1.020201	0.001316344	0.0013
0.3	1.041741	1.046028	0.004286584	0.0043
0.4	1.074303	1.083287	0.008983612	0.009
0.5	1.117479	1.133148	0.015668987	0.0157
0.6	1.172494	1.197217	0.024723507	0.0247
0.7	1.240946	1.277621	0.036675275	0.0367
0.8	1.324888	1.377128	0.052239484	0.0522
0.9	1.426929	1.499303	0.072373718	0.0724
1	1.550367	1.648721	0.098354149	0.0984
1.1	1.699372	1.831252	0.131880247	0.1319
1.2	1.879214	2.054433	0.175218806	0.1752
1.3	2.096575	2.327978	0.231402602	0.2314
1.4	2.359951	2.664456	0.304505463	0.3045
1.5	2.680192	3.080217	0.400024938	0.4
1.6	3.071222	3.59664	0.525417318	0.5254
1.7	3.551002	4.241852	0.690849732	0.6908
1.8	4.142827	5.05309	0.910263257	0.9103
1.9	4.877087	6.079971	1.202884333	1.2029
2	5.79367	7.389056	1.595386165	1.5954

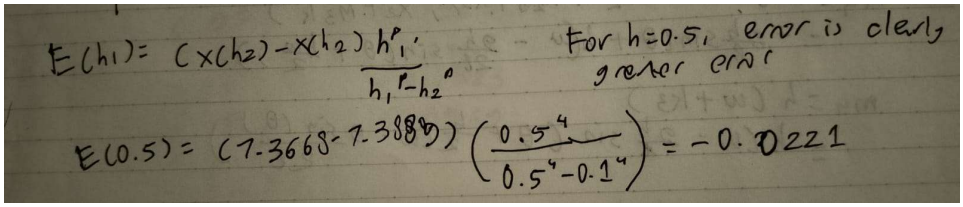
Here, the smallest and only negative error obtained is when t=0.1

When h=0.5

time	h0.5	Exact solution	Error	Error4digit	Runge-Kutta method
0	1	1	0	0	
0.5	1.149434	1.133148	−0.01629	−0.0163	
1	1.699568	1.648721	−0.05085	−0.0508	
1.5	3.290587	3.080217	−0.21037	−0.2104	
2	8.471662	7.389056	−1.08261	−1.0826	

For h=0.5, error is clearly greater error

As in section 4.7 Eq. (4-16), The h dependency between h=0.5 and h=0.1 is



(4-4) Similar to the Euler Method, the error in x(t) decreases as the value of h gets smaller. However, the Range-Kutta method has less error and thus comparatively more accurate answers for the same h as compared to the Euler method

time	h0.5	Exact solution	Error	Error4digit	Runge-Kutta method
0	1	1	0	0	
0.5	1.149434	1.133148	-0.01629	-0.0163	
1	1.699568	1.648721	-0.05085	-0.0508	
1.5	3.290587	3.080217	-0.21037	-0.2104	
2	8.471662	7.389056	-1.08261	-1.0826	

This is shown for the same h=0.5 value, Euler method:

time	h0.5	Exact solution	Error	Error4digit	Euler method
0	1	1	0	0	
0.5	1	1.133148	0.133148	0.1331	
1	1.25	1.648721	0.398721	0.3987	
1.5	1.875	3.080217	1.205217	1.2052	
2	3.28125	7.389056	4.107806	4.1078	

For the improved Euler Method

time	h0.5	Exact solu	Error	Error4digits	Improved euler method
0	1	1	0	0	
0.5	1.125	1.133148	0.008148	0.0081	
1	1.617188	1.648721	0.031534	0.0315	
1.5	2.931152	3.080217	0.149065	0.1491	
2	6.595093	7.389056	0.793963	0.794	

Runge-kutta method could have less error due to the existence of several parameters including k1, k2, k3, and k4 as opposed to just one. Also, Euler's method is first-order, so error reduces in magnitude with Δh. Improved Euler's method is second-order so error reduces with Δh^2. Runge-Kutta method is fourth-order, so error reduces in magnitude of Δh^4.

4.7 Numerical calculation

Finding the solution of the constant coefficient second-order ordinary differential equation of the pendulum shown b Eq. (2.17) $\ddot{\theta} + \omega_0{}^2 sin\theta = 0$

Task 7

(7-1) The ten equations in Eq. (4-14) are as the following

$$k_1 = h(-\frac{G}{L}sin(\theta_i))$$

$$m_1 = h \cdot \omega_i$$

$$k_2 = h(-\frac{G}{L}sin(\theta_i + \frac{m_1}{2}))$$

$$m_2 = h(\omega_i + \frac{k_1}{2}$$

$$k_3 = h(-\frac{G}{L}sin(\theta_i + \frac{m_2}{2}))$$

$$m_3 = h(\omega_i + \frac{k_2}{2}$$

$$k_4 = h(-\frac{G}{L}sin(\theta_i + m_3))$$

$$m_4 = h(\omega_i + k_3$$

$$\omega_{i+1} = \omega_i + \frac{k_1 + 2k_2 + 2k_3 + k4}{6}$$

$$\theta_{i+1} = \theta_i + \frac{m_1 + 2m_2 + 2m_3 + m4}{6}$$

(7-2) The code is modified as the following

We initially used h=0.001 and theta= 30/180*np.pi

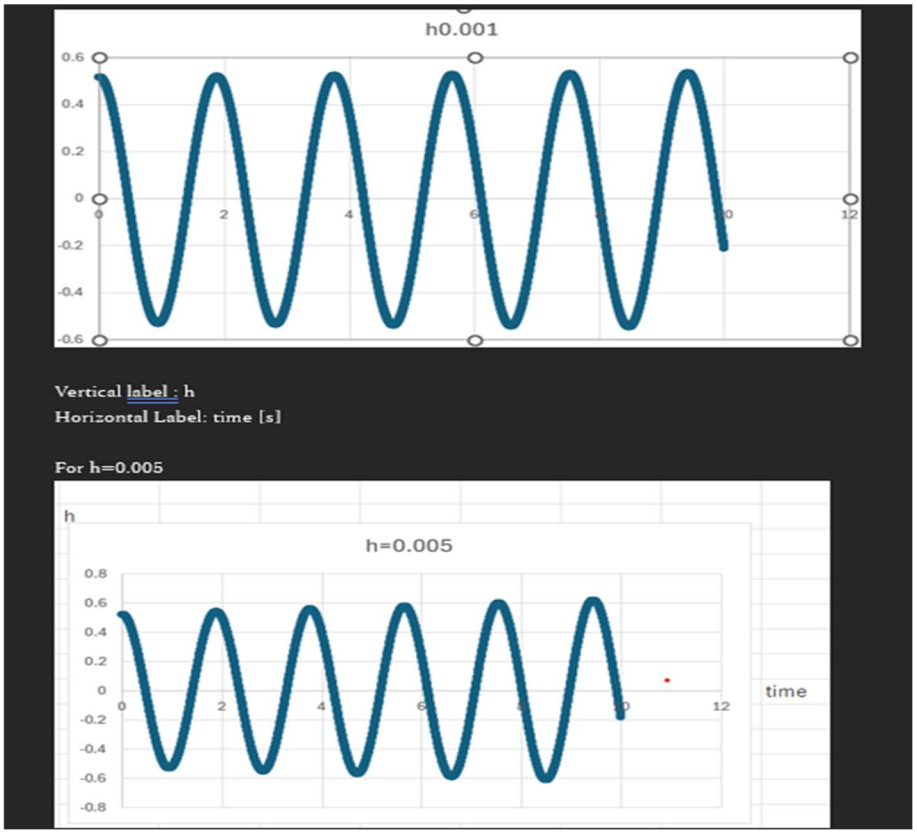
Line of code(lines 1-7 of code is just comments)

```
8   import numpy as np
9   #end time(s)
10  t_end=10
11  #time increments (change this)
12  h=0.005
13  #points
14  N=int(t_end/h+1)
15  #gravitational acceleration
16  G=9.7646 #rewrite
17  #pendulum length
18  L=0.705 #rewrite
19
20  #Times of Day
21  t=np.linspace(0,t_end,N)
22  #angular velocity (rad/s)
23  omega=np.linspace(0,t_end,N)
24  #Angle(rad)
25  theta= np.linspace(0,t_end,N)
26
27  #initial conditions (this part will be changed later)
28  omega[0] =0
29  theta[0] =30/180*np.pi #rad
30
31
32  #Runge-Kutta Method
33
34  for i in range(N-1):
35      k1= h* (-G/L*np.sin(theta[i]))
36      m1= h*omega[i]
37      k2= h* (-G/L*np.sin(theta[i])+m1/2)
38      m2= h*(omega[i] +k1/2)
39      k3= h* (-G/L*np.sin(theta[i])+m2/2)
40      m3=h*(omega[i]+k2/2)
```

```
        k4=h*(-G/L*np.sin(theta[i])+m3)
        d_omega=(k1+2*k2+2*k3+k4)/6
        d_theta= (m1+2*m2+2*m3)/6
        omega[i+1]=omega[i]+d_omega
        theta[i+1]= theta[i] +d_theta

with open('pendulum_RungeKutta.csv', 'w', newline = '')as f:
        f.write('time,h' +str(h))
        f.write('\n')
        for i in range(N):
            f.write(str(t[i])+',' +str(theta[i]))
            f.write('\n')
```

We then obtain



(7-4) Calculated angle at time t = 10 for different “h” values (Range-Kutta Method)

time	h0.001	h0.005
0	0.523599	0.523599
10	-0.20483	-0.20681

Task 8

(8-1 and 8-2)

In order to obtain the period of pendulum cycle from the calculation results of t and $\theta(t)$ using the Runge-Kutta method, we should change the value of h in the program such that the results should be equal to the exact solution to the 4 significant digits. And from here, we now that h = 0.001 should give the answer we are looking for.
Now from the value of h we have found, we changed the value of swing angle to other angles $\alpha = 5^o, 10^o, 20^o, 40^o$ by changing the theta[0] = α/180*np.pi in the program.
By doing so, we obtain the results of the theoretical value of the cycle period of respective swing angles:

Theoretical periods (s)

Swing angle	Period (s)
5°	1.689
10°	1.691
20°	1.701
30°	1.718
40°	1.741

Measured periods:

	5 degree	10 degree	20 degree	30 degree	40 degree
avg	1.68625	1.68325	1.70625	1.71125	1.73325

The slight dfference is between margin of error.

Theoritical method in Day 2 is more accurate, whereas experimentl method in Day 1 is more susceptible to random and measurement errors.